ROUNAK MUKHOPADHYAY

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SUMMARY

Research-oriented machine learning practitioner currently working on text-based deepfake detection and large language model behaviour analysis at IISER Bhopal. My work focuses on multilingual NLP, robustness under distribution shift, adversarial detection, and reliability of generative models in real-world settings. I am particularly interested in the security, evaluation, and interpretability of LLM-based systems, including model calibration and detection of compromised or synthetic outputs.

Research Focus: Reliability and robustness of generative models, distribution shift, multilingual NLP, and evaluation of large language systems in high-stakes settings.

EXPERIENCE

Project Associate-I – Machine Learning / NLP (Deepfake Detection) Jan 2026 – Present
Indian Institute of Science Education and Research (IISER) Bhopal
Project: Unmasking Digital Truth – Combating Deepfakes in Audio, Images, and Text for Enhanced Cybersecurity (ANRF, PMECRG)

- Investigating distributional and representational differences between human- and LLM-generated text in multilingual (Indic) settings.
- Designing transformer-based classification frameworks for detecting machine-generated text under low-resource and adversarial conditions.
- Evaluating robustness of detection models against paraphrasing, style-transfer, and prompt-based obfuscation attacks.
- Curating multilingual corpora and conducting systematic evaluation across cross-domain and cross-lingual generalisation settings.
- Analysing model confidence, calibration behaviour, and failure modes in high-dimensional semantic embedding spaces.

Project Associate – Human-Centered Machine Learning Dec 2024 – Dec 2025
BITS Pilani K. K. Birla Goa Campus — Sponsored by AI4ICPS, IIT Kharagpur
e-Monitor: AI System for Detecting Low Student Engagement in Online Learning

- Investigated generalisation of multimodal engagement detection models across users under distributional variability and behavioural heterogeneity.
- Designed multimodal learning frameworks integrating physiological (GSR, HR) and ocular signals to model cognitive engagement using feature-engineered and ensemble-based approaches.
- Studied robustness under sensor noise, signal dropout, and missing data; analysed model calibration and confidence behaviour in deployment-like settings.
- Explored active learning and personalised calibration strategies to improve cross-user reliability while reducing manual annotation requirements.
- Evaluated model performance using accuracy and macro F1 under subject-wise validation to prevent leakage and overfitting.

Research Intern – Bioinformatics Jan 2024 – Jun 2024
University of Kalyani, West Bengal
In-silico identification and characterisation of virus-associated circular RNAs

- Conducted computational analysis of RNA-seq datasets to identify and characterise circRNAs potentially associated with virus-induced cancers.

- Automated alignment, annotation, and reporting workflows using Python and R to improve reproducibility and reduce processing time.

RESEARCH PROJECTS

Multi-task MRI-based Lumbar Spine Degeneration Classification (RSNA 2024)

Developed a multi-task deep learning framework to predict degenerative severity (Normal/Mild, Moderate, Severe) across multiple anatomical condition-level pairs from lumbar spine MRI scans.

- Curated and structured raw DICOM datasets (study → series → slices), aligned radiological labels with MRI sequence metadata (Sagittal T1/T2, Axial T2), and formalised the task as a multi-condition classification problem with shared representation learning.
- Implemented a ResNet18-based multi-task architecture with condition-specific output heads; trained using cross-entropy across condition-level pairs and evaluated using accuracy and macro F1 to account for class imbalance.
- Achieved 78% accuracy under patient-level stratified validation; analysed performance degradation due to imbalance (macro F1 ≈ 0.41) and investigated weighted loss strategies for robustness.
- Explored architectural extensions including multi-slice (2.5D) context aggregation and multi-sequence fusion to improve anatomical representation learning.
- Evaluated generalisation under limited data conditions and examined calibration behaviour to assess reliability in clinical deployment scenarios.

Multimodal Emotion Inference

Designed machine learning models to infer affective states from physiological and ocular signals; evaluated generalisation under cross-session variability and missing signal conditions.

RESEARCH AND TECHNICAL EXPERTISE

Machine Learning & NLP: Transformer-based language models, representation learning in high-dimensional semantic spaces, multilingual text modelling, generative model detection, robustness under distribution shift, model calibration and uncertainty analysis

Deep Learning: Multi-task learning, 2D/3D CNNs for medical imaging, multimodal modelling, active learning, class-imbalance mitigation (weighted loss, focal loss)

Programming & ML Frameworks: Python (primary); PyTorch, TensorFlow, Scikit-learn

Scientific Computing: NumPy, Pandas, Matplotlib

Infrastructure & Deployment: Linux-based workflows; Git version control; Flask-based API deployment

Databases: PostgreSQL, MySQL

EDUCATION

B.Sc. Data Science and Applications 2022 – Present
Indian Institute of Technology Madras
CGPA: 7.1/10 (ongoing)

M.Sc. in Bioinformatics 2022 – 2024
Maulana Abul Kalam Azad University of Technology
CGPA: 7.94/10

B.Sc. (Hons.) Microbiology 2019 – 2022
West Bengal State University
CGPA: 9.19/10